Title:

Predictive Forecasting of Care Load and Placement Demand Using Machine Learning

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Abstract

This study focuses on predicting future care load in the Unaccompanied Alien Children (UAC) program using machine learning techniques. By analyzing historical time-series data, the project aims to provide short-term forecasts that help in proactive decision-making. A Random Forest model is implemented to generate predictions based on previous trends, ensuring reliability even with incomplete datasets.

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1. Introduction

The UAC program operates in a highly uncertain environment where sudden changes can increase the number of children under care. Traditional analytics only explain past data, but this project aims to predict future demand using machine learning.

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2. Objectives

- Forecast children in HHS care
- Identify future demand trends
- Enable proactive resource planning

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3. Dataset Description

The dataset includes:

- Date
- Children in HHS Care
- Children transferred
- Children discharged

The data required preprocessing due to:

- Comma-separated numeric values
- Missing values

- Date formatting issues

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4. Methodology

Data Cleaning:

- Removed commas from numeric values
- Converted columns to numeric
- Handled missing values using forward/backward fill

Feature Engineering:

- Lag feature (previous day value)

Model Used:

- Random Forest Regressor

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5. Model Implementation

The model predicts future care load based on historical patterns using lag-based features.

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6. Results

- Accurate short-term forecasting
- Stable predictions even with incomplete data
- Effective for real-world deployment

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7. Conclusion

The project successfully transforms historical data into predictive intelligence. It enables policymakers to anticipate future care demand and allocate resources efficiently.

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8. Future Scope

- Add ARIMA / advanced models
- Include seasonal trends

- Improve accuracy with more features